

Accuracy of Sentinel Lymph Node Biopsy to Predict Nodal Metastasis in Head and Neck Merkel Cell Carcinoma

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[+ Supplemental content](#)

IMPORTANCE Merkel cell carcinoma (MCC) is a rare neuroendocrine malignant neoplasm often affecting the head and neck. Sentinel lymph node biopsy (SLNB) is routinely performed; however, the accuracy and reliability of SLNB for head and neck MCC remains unknown.

OBJECTIVE To determine the accuracy of SLNB in head and neck MCC and inform pre- and post-SLNB risk stratification.

DESIGN, SETTING, AND PARTICIPANTS This was a single-center cohort study of patients with clinically node-negative head and neck MCC who underwent SLNB from January 2006 to January 2025; median (IQR) follow-up was 1.9 (0.8-4.6) years. Data analysis was conducted from April to November 2025.

INTERVENTIONS SLNB outcome and nodal disease status.

MAIN OUTCOMES AND MEASURES SLNB accuracy measured by sensitivity, negative predictive value (NPV), and false negative (FN) rate. FN SLNB was defined as regional nodal recurrence at the first site of recurrence after a negative SLNB without concurrent local or in-transit recurrence. Factors associated with failed SLNB (defined as nonvisualization on lymphoscintigraphy, inability to identify the sentinel lymph node intraoperatively, or absence of nodal tissue in the pathology specimen) were evaluated using effect-size measures. Cox proportional hazards model identified factors associated with nodal disease, comparing patients who developed nodal disease (identified by SLNB or subsequent clinical and/or imaging evaluation) with those who did not develop nodal disease during follow-up.

RESULTS Of 86 participants (mean [SD] age, 75.6 [9.6] years; 36 female [41.9%] and 50 male [58.1%] individuals) with clinically node-negative head and neck MCC, 69 (80.2%) underwent SLNB. Occult nodal disease was identified in 19 participants (27.5%). SLNB sensitivity was 52.6% (10 of 19 patients), NPV was 88.6% (31 of 35 patients), and FN rate was 26.7% (4 of 15 patients). SLNB failed in 29.0% (20 of 69 patients), with tumor location, female sex, and older age having medium to large effect on SLNB failure. Lymphovascular invasion, infiltrative growth pattern, and tumor size had the largest effect size on nodal disease with Cox analysis.

CONCLUSIONS AND RELEVANCE In this large single-site cohort study, SLNB remained a useful tool in head and neck MCC, identifying more than half of occult nodal disease. Given reduced performance in this region, patients with failed SLNB or negative SLNB plus high-risk features may benefit from tailored management, including intensified therapy and surveillance.

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Merkel cell carcinoma (MCC) is a rare and aggressive neuroendocrine skin cancer with a high risk for nodal metastasis and recurrence.¹ Sentinel lymph node biopsy (SLNB) is critical for staging, management, and prognostication, and the US National Comprehensive Cancer Network (NCCN) guidelines recommend SLNB for all patients who are medically fit and have clinically node-negative disease.²

Between 34% and 44% of MCC cases arise in the head and neck region.^{1,3-6} This location presents unique challenges for SLNB due to complex lymphatic drainage patterns and proximity to critical anatomical structures.^{7,8} Accurate nodal staging is especially important because head and neck MCCs (HN MCC) may have worse prognoses than non-HN MCC.^{9,10}

Prior studies of SLNB and HN MCC have been limited by small sample sizes (often <10 patients) and have reported relatively high false-negative rates.^{8,11-13} A multisite retrospective study¹⁴ characterizing predictors of FN SLNB in stage I/II MCC included 214 patients with HN MCC and found a 39% FN rate in the HN cohort but did not otherwise analyze this cohort separately or describe the unique considerations associated with this region. High SLNB failure rates have also been observed in HN melanoma and squamous cell carcinoma and been largely attributed to anatomic complexity.¹⁵⁻¹⁷

In this large cohort study of patients with clinically node-negative HN MCC who underwent SLNB, we sought to evaluate SLNB accuracy within this anatomically complex region and to identify factors associated with SLNB failure and nodal disease. Our goal was to inform staging practices and refine risk stratification strategies to improve treatment planning and outcomes for this high-risk population.

Methods

This retrospective study was approved by the institutional review board of the University of Pennsylvania (protocol #842985), which granted a waiver of informed consent due to the minimal risk nature of the study and the use of existing clinical data. We followed the Strengthening of Reporting of Observational Studies in Epidemiology (STROBE) reporting guidelines.

Key Points

Question What is the accuracy of sentinel lymph node biopsy (SLNB) in head and neck Merkel cell carcinoma?

Findings This single-center retrospective cohort study of 69 patients with clinically node-negative head and neck Merkel cell carcinoma who underwent SLNB found that occult nodal disease was present in 19 patients (27.5%). SLNB sensitivity was 52.6% (10 of 19 patients), with high rates of failed SLNB (29.0%; 20 of 69 patients) and false negative SLNB (26.7%; 4 of 15 patients).

Meaning These findings suggest that patients with failed SLNB or those with negative SLNB but high-risk features may benefit from tailored therapy and surveillance strategies.

Study Design and Setting

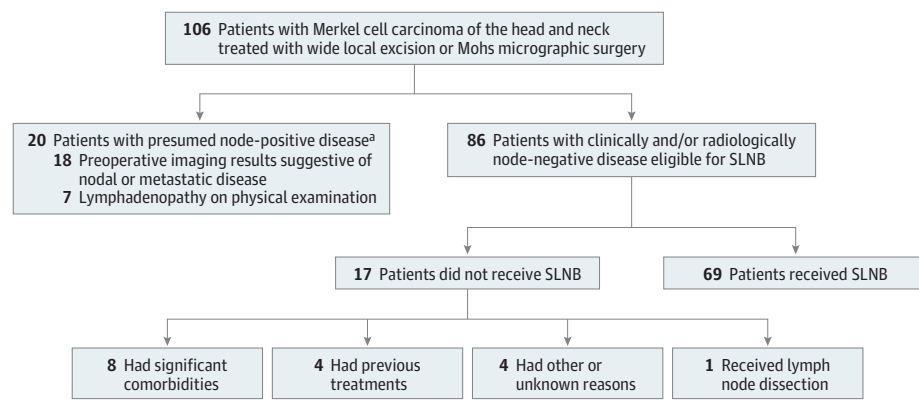
Histologically confirmed cases of HN MCC treated with Mohs micrographic surgery (MMS) or wide local excision (WLE) between January 2006 and January 2025 were identified from prospectively maintained databases. Patients with lymphadenopathy on physical examination or radiologic evidence of nodal or metastatic disease before surgery were excluded (Figure 1).

At our institution, SLNB involves initial mapping with technetium-99m lymphoscintigraphy, followed by intra-operative localization using a gamma probe. Blue dye (isosulfan blue or methylene blue) is used at surgeon discretion. Sentinel lymph nodes were serially sectioned and examined with hematoxylin and eosin at multiple levels. If metastatic carcinoma was not identified, immunohistochemical staining for cytokeratin (using the monoclonal antibody combination AE1/AE3 and/or cytokeratin 20) was performed to evaluate for micrometastatic Merkel cell carcinoma. In cases with equivocal morphologic or immunophenotypic findings, additional neuroendocrine markers (eg, synaptophysin and/or chromogranin) were applied for further characterization.

Data Collection

Data were extracted from electronic medical records using REDCap, version 15.9.3, and included demographic character-

Figure 1. Flow Diagram of Participants Who Underwent Sentinel Lymph Node Biopsy (SLNB)



^aSome patients had findings from both preoperative imaging and physical examination that were suggestive of nodal or metastatic disease; therefore, the sum is greater than 20.

istics, smoking status, immunosuppression, tumor features, surgical characteristics, cancer stage (per the *American Joint Committee on Cancer Staging Manual, eighth edition*), SLNB performance, imaging, adjuvant treatment, recurrence, and survival. Patients were considered immunosuppressed if they had undergone organ transplant, had autoimmune disease, hematologic malignant disease (eg, myeloma, polycythemia vera), and/or were receiving long-term immunosuppressive therapy.

Tumor diameter was taken from pathology reports or, if not reported, from operative or biopsy specimen documentation. Recorded pathologic features included tumor-infiltrating lymphocytes, lymphovascular invasion (LVI), mitoses, infiltrative growth pattern, ulceration and/or necrosis, and presence of a second cancer type in the specimen. High mitotic rate was defined as 10 or more mitotic figures per high-power field.¹⁸

Location of biopsied sentinel lymph node (SLN) was determined from surgeon operative notes, referencing lymphoscintigraphy or pathology reports when needed. SLN location was categorized as parotid, perifacial, upper neck (levels I-III), lower neck (level IV), or posterior neck and/or scalp (including level V).

Study Cohort

A total of 106 patients with HN MCC treated with WLE or MMS at our institution were identified, 86 of whom had no clinical or radiologic evidence of nodal or metastatic disease before surgery. Of these, 69 patients (80.2%) underwent SLNB (Figure 1). Only 1 patient lacked documented discussion of SLNB and 17 (19.8%) did not undergo SLNB: 3 due to prior excision and closure at an outside institution thought to have disrupted lymphatics, 8 due to comorbidities, 4 who declined for unknown reasons, 1 with same-day biopsy and MMS followed by imaging showing nodal disease, and 1 who underwent completion lymph node dissection due to high-risk features.

Classification of SLNB

SLNB results were categorized as true positive (TP), true negative (TN), false negative (FN), or failed. A TP was defined as an SLNB that detected metastatic disease. A TN was defined as an SLNB that showed no disease and was not followed by regional nodal recurrence during the follow-up period.¹⁴ SLNB was still considered TN if the patient developed local, in-transit, or distant recurrence. An FN SLNB was defined as regional nodal recurrence as the first site of recurrence without concurrent local or in-transit recurrence.^{7,17,19,20} A failed SLNB was defined as inability to obtain a SLN for evaluation, either due to nonvisualization on lymphoscintigraphy, inability to identify the SLN intraoperatively, absence of nodal tissue in the pathology specimen, or failure to inject radiotracer. Time to nodal recurrence was calculated from the SLNB date to the date of pathologic confirmation of recurrence. If pathologic confirmation was unavailable, the date of confirmatory imaging was used. Follow-up period was measured from SLNB to patient's last in-office visit with a clinician treating the MCC (eg, oncology, dermatology, otolaryngology).

Primary and Secondary Outcomes

The primary outcome was SLNB accuracy, measured by sensitivity, negative predictive value (NPV), and FN rate. Patients with negative or failed SLNB who received adjuvant radiation therapy (aRT) to the draining nodal basin were excluded from accuracy calculations, but those receiving aRT to the primary site alone were included.¹⁴ Patients with nodal disease concurrent with local or in-transit recurrence were excluded from accuracy calculations. Secondary outcomes included factors associated with failed SLNB and predictors of nodal disease among patients who are clinically node negative.

Statistical Analysis

Sensitivity was calculated as (number of TP ÷ [number of TP + number of FN + number of patients with failed SLNB with nodal disease]). NPV was calculated as (number of TN ÷ [number of TN + number of FN]). FN rate was calculated as (number of FN ÷ [number of FN + number of TP]). Specificity and positive predictive value were not calculated because all patients with positive SLNB were assumed to have nodal disease.

Effect-size measures with 95% CI were used to characterize predictors of failed SLNB. Mean difference and standardized mean difference with Cohen *d* were used for continuous variables (eg, age), with a small effect suggested when *d* value was 0.2 to less than 0.5; medium effect, when 0.5 to less than 0.8; and large effect, when 0.8 or greater. Percent difference was used for binary variables (eg, sex).²¹ Cramer V was reported for nominal variables with 3 or more groups, with a negligible relationship suggested if less than 0.10, a small relationship when 0.10 to less than 0.30, a medium relationship when 0.30 to less than 0.50, and a large relationship when 0.50 or greater.^{21,22} Cliff δ was used for ordinal variables with 3 or more groups, with a negligible relationship suggested if less than 0.15, a small relationship if 0.15 to less than 0.33, a medium effect if 0.33 and 0.47, and a large relationship if 0.47 or greater.²³ A supplemental bivariable analysis for predictors of failed SLNB was also conducted for categorical variables. To account for follow-up time, univariable Cox proportional hazards models identified factors associated with nodal disease.

Using the study's clinically node-negative cohort that underwent SLNB, we compared patients without nodal disease who did not receive aRT to the draining nodal basin to patients ultimately found to have nodal disease (ie, patients with positive SLNB, patients with FN SLNB, and patients with failed SLNB found to have nodal disease through postsurgery physical examination and/or imaging without concurrent local or in-transit recurrence). Data analysis was performed from April to November 2025 using R Studio, version 2024.04.02 (R Foundation for Statistical Computing).

Results

Of 86 participants (mean [SD] age, 75.6 [9.6] years; 36 female [41.9%] and 50 male [58.1%] individuals) with clinically node-negative HN MCC, 69 (80.2%) underwent SLNB; additional characteristics are available in Table 1. Patients were most com-

Table 1. Characteristics of Patients With Clinically Node-Negative Head and Neck Merkel Cell Carcinoma Who Underwent Sentinel Lymph Node Biopsy (SLNB), by Success or Failure

Characteristic	No. (%)			Effect size, % (95% CI) ^a
	Participants	Success	Failure	
Participants	69 (100)	49 (71)	20 (29)	NA
Sex				
Female	28 (40.6)	16 (32.7)	12 (60.0)	-27.3 (-56.0 to 1.3)
Male [Reference]	41 (59.4)	33 (67.3)	8 (40.0)	
Race (EMR data)				
White [Reference]	66 (95.7)	46 (93.9)	20 (100)	6.1 (-4.1 to 16.4)
Not White	3 (4.3)	3 (6.1)	0	
Age at surgery, median (IQR), y ^b	75 (69-81)	73 (68-80)	78 (72-84)	-4.9 (-9.2 to -0.56)
Smoking status				
No	37 (53.6)	29 (59.2)	8 (40.0)	0.18 (0 to 0.42)
Former	26 (37.7)	16 (32.7)	10 (50.0)	
Yes	6 (8.7)	4 (8.2)	2 (10.0)	
Immunosuppressed				
No [Reference]	58 (84.1)	40 (81.6)	18 (90.0)	8.4 (-12.2 to 28.9)
Yes	11 (15.9)	9 (18.4)	2 (10.0)	
Use of SPECT/CT				
No [Reference]	57 (82.6)	41 (83.7)	16 (80.0)	-3.7 (-27.6 to 20.2)
Yes	12 (17.4)	8 (16.3)	4 (20.0)	
Year of SLNB				
2010-2019 [Reference]	32 (46.4)	23 (46.9)	9 (45.0)	-1.9 (-30.0 to 26.0)
2020-2025	37 (53.6)	26 (53.1)	11 (55.0)	
Previous excisional biopsy				
No [Reference]	57 (82.6)	41 (83.7)	16 (80.0)	-3.7 (-27.6 to 20.2)
Yes	12 (17.4)	8 (16.3)	4 (20.0)	
T stage				
pT1	64 (92.8)	47 (95.9)	17 (85.0)	0.11 (-0.07 to 0.28)
pT2	3 (4.3)	1 (2.0)	2 (10.0)	
pT3	2 (2.9)	1 (2.0)	1 (5.0)	
Tumor diameter, median (IQR), cm ^c	0.90 (0.60 to 1.40)	1.00 (0.65)	1.33 (1.21)	-0.33 (-0.92 to 0.25)
Tumor thickness, median (IQR), mm ^d	2.75 (1.55 to 5.15)	3.52 (2.54)	4.51 (4.24)	-0.99 (-3.5 to 1.50)
Unknown, No.	25	20	5	NA
Tumor thickness, mm				
<1	5 (11.4)	4 of 29 (13.8)	1 of 15 (6.7)	0.19 (-0.14 to 0.48)
1 to <5	28 (63.6)	19 of 29 (65.5)	9 of 15 (60.0)	
5 to <10	9 (20.5)	6 of 29 (20.7)	3 of 15 (20.0)	
≥10	2 (4.5)	0	2 of 15 (13.3)	
Unknown, No.	25	20	5	
Tumor location				
Ear	6 (8.7)	6 (12.2)	0	0.55 (0.38 to 0.78)
Temple	4 (5.8)	3 (6.1)	1 (5.0)	
Periocular	7 (10.1)	5 (10.2)	2 (10.0)	
Cheek	18 (26.1)	8 (16.3)	10 (50.0)	
Scalp	5 (7.2)	4 (8.2)	1 (5.0)	
Nose	10 (14.5)	10.0 (20.4)	0	
Forehead	8 (11.6)	7 (14.3)	1 (5.0)	
Chin	4 (5.8)	3 (6.1)	1 (5.0)	
Neck	5 (7.2)	1 (2.0)	4 (20.0)	
Perioral or lip	2 (2.9)	2 (4.0)	0	

(continued)

Table 1. Characteristics of Patients With Clinically Node-Negative Head and Neck Merkel Cell Carcinoma Who Underwent Sentinel Lymph Node Biopsy (SLNB), by Success or Failure (continued)

Characteristic	No. (%)		SLNB status		Effect size, % (95% CI) ^a
	Participants		Success	Failure	
Surgery type					
MMS [Reference]	55 (79.7)		39 (79.6)	16 (80.0)	0.41 (−20.8 to 21.7)
WLE	14 (20.3)		10 (20.4)	4 (20.0)	
Lymphovascular invasion	21 (30.4)		12 (24.5)	9 (45.0)	−20.5 (−48.9 to 7.9)
Tumor infiltrating lymphocytes	28 (40.6)		18 (36.7)	10 (50.0)	−13.3 (−42.5 to 16.0)
Mitoses					
None noted	7 (15.9)		4/28 (14.3)	3/16 (18.8)	
Low	9 (20.5)		8/28 (28.6)	1/16 (6.2)	0.18 (−0.14 to 0.46)
High	28 (63.6)		16/28 (57.1)	12/16 (75.0)	
Unknown, No.	25		21	4	
Infiltrative growth pattern	10 (14.5)		6 (12.2)	4 (20.0)	−7.8 (−31.1 to 15.6)
Ulceration or necrosis	27 (39.1)		22 (44.9)	5 (25.0)	20.0 (−7.2 to 47.0)
Second malignant disease in pathologic section	8 (11.6)		5 (10.2)	3 (15.0)	−4.8 (−26.1 to 16.5)
Invasion into bone, cartilage, fascia, or muscle	11 (15.9)		6 (12.2)	5 (25.0)	−12.8 (−37.4 to 11.8)

Abbreviations: EMR, electronic medical record; MMS, Mohs micrographic surgery; NA, not applicable; SPECT/CT, single photon emission computed tomography/computed tomography; WLE, wide local excision.

^a Effect size (95% CI) is displayed for each factor to describe its predictive impact on failed SLNB; standardized mean difference (Cohen *d* [95% CI]) for continuous variables appear in corresponding footnotes; percent difference (95% CI) for binary variables; Cramer *V* (95% CI) for nominal variables with 3

or more categories; and Cliff δ (95% CI) for ordinal variables with 3 or more categories.

^b Cohen *d* (95% CI), −0.59 (−1.1 to −0.06) y.

^c Cohen *d* (95% CI), −0.40 (−0.92 to 0.13) cm.

^d Cohen *d* (95% CI), −0.31 (−0.93 to 0.32) mm.

monly treated with MMS (55 [78.6%]). Almost all patients (68 [98.6%]) received SLNB either before or at the same time as their definitive primary site surgery. Twelve patients (17.4%) underwent a diagnostic excisional biopsy procedure of the primary tumor before SLNB.

Among patients with successful SLNB, 90 nodes were removed (mean [SD], 1.84 [1.51] nodes per patient). SLN location was most often in the parotid region (24 of 48 patients [50.0%]), followed by upper cervical (16 of 44 patients [33.3%]), perioral (4 of 48 patients [8.3%]), lower cervical (2 of 48 patients [4.2%]), and posterior cervical (2 of 48 patients [4.2%]).

Blue dye was used in 38 of the 52 patients (73.1%) with available data. Single photon emission computed tomography/computed tomography (SPECT/CT) was used in 12 patients (mean [SD] of 2.25 [1.75] nodes removed per patient when SLNB was successful). Median (IQR) follow-up was 1.9 (0.8–4.5) years.

Accuracy of SLNB

Figure 2 displays the SLNB outcomes. Of the 69 patients who underwent SLNB, 20 procedures (29.0%) failed, 11 (15.9%) were positive, and 38 (55.1%) were negative. Nine patients with either negative or failed SLNB received aRT to the nodal basin. Two patients with failed SLNB underwent completion lymph node dissection: 1 had negative and 1 had positive results for nodal disease.

Among 35 patients with negative SNLB and no aRT to the nodal basin, 4 (11.4%) developed nodal disease as the first site of recurrence without local or in-transit recurrence at a mean (SD) of 260 (190.6) days after SLNB (median [IQR], 198.5 [119.0–526.0]) days. All 4 recurrences were detected by palpable

lymphadenopathy—2 in the same nodal group as SLNB (parotid) and 2 in new neck levels. FN SLNB, defined as regional nodal recurrence as first site of recurrence after prior negative SLNB, occurred in 4 patients. The FN rate was 26.7% (4 of 15 patients) and NPV was 88.6% (31 of 35 patients).

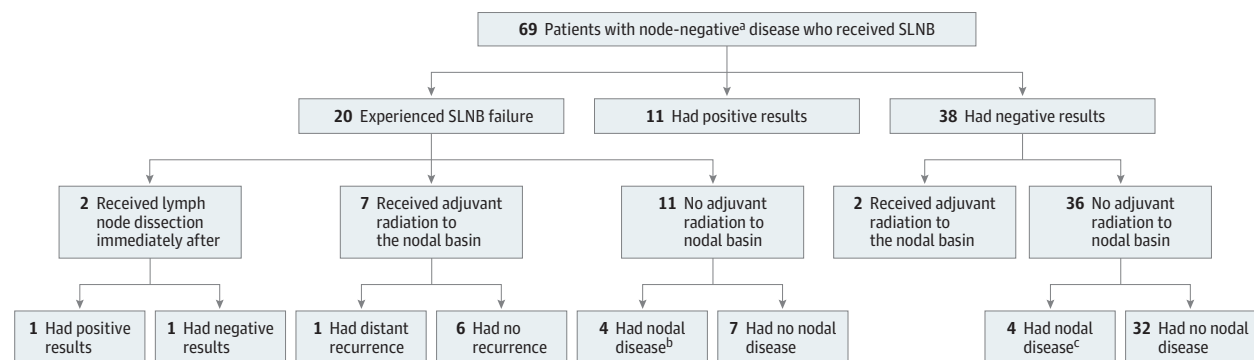
Among 11 patients who experienced failed SLNB and no aRT, 4 (36.3%) developed nodal disease, detected a mean (SD) of 102.7 (73.4) days after SLNB (median [IQR] days, 124 [31–197]), all by imaging. Based on positive, negative, and failed SLNB outcomes, overall SLNB sensitivity was 52.6% (10 of 19 patients).

SLNB Failure

In the 69 patients undergoing SLNB, 20 of these procedures failed (29.0%). Reasons included nonvisualization in 13 cases (65.0%), inability to identify the SLN intraoperatively in 4 (20.0%), absence of nodal tissue in the pathology specimen in 2 (10.0%), and inability to inject radiotracer in 1 (5.0%). Predictors of SLNB failure are shown in Table 1 and the eTable in Supplement 1.

Tumor location had a large effect on SLNB failure (Cramer *V* = 0.55; 95% CI, 0.38–0.78). Primary tumors on the cheek and neck had failure rates greater than 50% (10 of 17 and 4 of 5 patients, respectively), while tumors on the nose and ear had a 100% success rate (10 of 10 and 8 of 8, respectively). Age (*d* = −0.59, 95% CI, −1.10 to −0.06) and sex (percentage difference, −27.3%; 95% CI, −56.0% to 1.3%; odds ratio [OR], 3.09; 95% CI, 1.07 to 9.39) both had a moderate effect on SLNB failure, with older patients and female patients experiencing higher rates of SLNB failure.

Figure 2. Outcomes for Patients With Clinically Node-Negative Head and Neck Merkel Cell Carcinoma Who Had Sentinel Lymph Node Biopsy (SLNB)



Additional surgical and/or adjuvant radiation therapy and subsequent development of nodal disease are included.

^aPatients with clinically and/or radiologically node-negative disease.

^bOne patient was found to have nodal disease at computed tomography–simu-

lation appointment with radiation oncology.

^cTwo patients had nodal disease in a different basin that was sampled during SLNB.

Other factors, such as immunosuppression, LVI, depth of invasion, and tumor size, had small or negligible effect on SLNB failure. Notably, 9 of 20 patients (45.0%) with LVI experienced failed SLNB. Among the 5 patients with tumors larger than 2 cm, 3 (60%) experienced SLNB failure. Additionally, 4 of 5 tumors measuring less than 1 mm (80%) were treated successfully with SLNB, while 2 tumors measuring more than 10 mm failed SLNB. Blue dye was used in 4 of the 6 failed intraoperative cases.

Predictors of Nodal Disease

On univariable Cox analysis, LVI (hazard ratio [HR], 4.87; 95% CI, 1.22-19.5), infiltrative growth pattern (HR, 4.63; 95% CI, 1.03-20.70), and tumor size (HR, 2.62; 95% CI, 1.20-5.71) had large effect size on nodal disease, while tumor thickness (HR, 1.39; 95% CI, 1.05-1.83) and age (HR, 1.12; 95% CI, 1.01-1.24) had smaller effect size on nodal disease (Table 2). Notably, 5 of 10 patients with invasion into deeper structures (eg, bone, cartilage, fascia, muscle) and 4 of 8 patients with a second malignant neoplasms or disease in the pathologic section had nodal disease. Mean (SD) follow-up period in the no-nodal disease group was 2.9 (3.0) years.

Discussion

In this large single center cohort study focusing on SLNB accuracy in clinically node-negative HN MCC. SLNB is an important component of staging, treatment planning, and prognostication, and National Comprehensive Cancer Network guidelines recommend it for all patients who are medically fit and have clinically node-negative disease.² However, in HN, its performance can be affected by complex and variable lymphatic drainage and overlapping nodal basins.^{7,8} Prior studies of HN MCC have reported relatively high FN rates, but have either been limited in size (often <10 patients), or lacked additional analysis specific to the HN

cohort.^{8,11-14} Our findings confirm a substantial burden of clinically occult nodal disease, with 27.5% of the cohort harboring disease undetected on physical examination or imaging, consistent with prior reports,^{4,8,14,24} and demonstrate that SLNB can identify more than half of these cases. At the same time, SLNB failure occurred in 29.0% of cases, with potentially higher risk in particular tumor locations, such as the cheek or neck, females, and older patients. LVI, tumor diameter, and tumor thickness were predictors of nodal disease, consistent with prior literature.⁴ These results underscore both the value of SLNB and the need for complementary strategies to optimize nodal staging in HN MCC.

In this study's cohort, SLNB sensitivity was 52.6%, NPV was 88.6%, and FN rate was 26.7%. Nodal disease was identified in 36.4% of patients with failed SLNB and no aRT to the nodal basin, compared with 11.4% of patients with negative SLNB and no aRT to the nodal basin. Notably, all nodal recurrences after failed SLNB were detected earlier (mean [SD] days, 102.7 [73.4]) and by imaging, while FN SLNB cases were detected later (mean [SD] days, 260.0 [190.6]) and by physical examination. This pattern suggests that follow-up imaging may be used more aggressively after failed SLNB and more sparingly after negative SLNB, and that imaging can detect subclinical nodal disease earlier.^{25,26} These data support consideration of early surveillance imaging, particularly within the first 3 to 6 months after SLNB for patients with failed SLNB or negative SLNB and high-risk features.

The FN rate we found (26.7%) aligns with smaller HN MCC studies but exceeds rates reported in larger multi-institutional series.^{8,12-14,24} Differences likely reflect variation in FN definitions; larger studies often count only recurrences in the same basin as FN,^{8,14,24} whereas we included new basin nodal recurrences in the absence of local or in-transit recurrence.^{7,17,19,20} High FN rates have also been reported in HN melanoma, with some series reporting rates greater than 30%.^{7,27,28}

Table 2. Cox Proportional Hazards Model Finding Predictors of Nodal Disease in Patients With Clinically Node-Negative Head and Neck Merkel Cell Carcinoma Who Underwent Sentinel Lymph Node Biopsy (SLNB)

Variable	Disease status ^a		HR (95% CI)
	No nodal disease	Nodal disease	
Patients, No.	40	19	NA
Age at surgery, y	74.5 (8.8)	74.8 (9.1)	1.12 (1.01-1.24)
Sex			
Female	20 (80.0)	5 (20.0)	1.16 (0.26-5.2)
Male	20 (58.8)	14 (41.2)	1.00 [Reference]
Smoking status			
Never	23 (65.7)	12 (34.3)	1.00 [Reference]
Former	14 (73.7)	5 (26.3)	0.91 (0.17-4.98)
Current	3 (60.0)	2 (40.0)	1.61 (0.18-14.46)
Immunosuppression status			
Competent	34 (69.4)	15 (30.6)	1.00 [Reference]
Suppressed	6 (60.0)	4 (40.0)	0.83 (0.10-6.88)
Prior treatment			
None	33 (66.0)	17 (34.0)	1.00 [Reference]
Prior excisional biopsy or WLE	7 (77.8)	2 (22.2)	0.65 (0.08-5.38)
Lymphovascular invasion			
Absent	33 (75.0)	9 (21.4)	1.00 [Reference]
Present	7 (41.2)	10 (58.8)	6.27 (1.4-28.1)
Tumor diameter, cm	0.89 (0.6)	1.44 (0.9)	2.60 (1.14-5.89)
Tumor thickness, mm	3.38 (2.47)	4.27 (3.86)	1.39 (1.05-1.83)
Tumor location			
Cheek	8 (61.5)	5 (38.5)	1.00 [Reference]
Temple	3 (100)	0	0 (0-∞)
Periocular	4 (80.0)	1 (20.0)	0 (0-∞)
Ear	4 (66.7)	2 (33.3)	0.72 (0.07-6.95)
Scalp	3 (75.0)	1 (25.0)	0 (0-∞)
Nose	8 (80.0)	2 (20.0)	0 (0-∞)
Forehead	5 (62.5)	3 (37.5)	0.68 (0.07-6.55)
Chin	3 (75.0)	1 (25.0)	0.98 (0.10-9.44)
Neck	1 (25.0)	3 (75.0)	2.15 (0.22-20.74)
Perioral or lip	1 (50.0)	1 (50.0)	0 (0-∞)
Surgery type			
MMS	33 (70.2)	14 (29.8)	1.00 [Reference]
WLE	7(58.3)	5 (41.7)	1.52 (0.30-7.85)
Tumor infiltrating lymphocytes			
None	24 (66.7)	11 (31.4)	1.00 [Reference]
Present	16 (66.7)	8 (33.3)	1.22 (0.27-5.47)
Mitoses			
None	3 (60.0)	2 (40.0)	1.00 [Reference]
Low	6 (75.0)	2 (25.0)	(0-∞)
High	13 (54.2)	11 (45.8)	0.47 (0.09-2.6)
Not specified	18 (81.8)	4 (18.2)	NA
Infiltrative growth pattern			
No	36 (72.0)	14 (28.0)	1.00 [Reference]
Yes	4 (44.4)	5 (55.6)	4.63 (1.03-20.7)
Ulceration or necrosis			
No	20 (60.6)	13 (39.4)	1.00 [Reference]
Yes	20 (76.9)	6 (23.1)	0.38 (0.07-1.95)
Second malignant disease within pathologic section			

(continued)

Table 2. Cox Proportional Hazards Model Finding Predictors of Nodal Disease in Patients With Clinically Node-Negative Head and Neck Merkel Cell Carcinoma Who Underwent Sentinel Lymph Node Biopsy (SLNB) (continued)

Variable	Disease status ^a		HR (95% CI)
	No nodal disease	Nodal disease	
No	36 (70.6)	16 (29.4)	1.00 [Reference]
Yes	4 (50.0)	4 (50.0)	3.45 (0.67-17.9)
Invasion into bone, cartilage, fascia, or muscle			
No	35 (70.0)	14 (28.6)	1.00 [Reference]
Yes	5 (50.0)	5 (50.0)	2.05 (0.40-10.6)

Abbreviations: HR, hazard ratio; MMS, Mohs micrographic surgery; NA, not applicable; WLE, wide local excision.

^a Patients were considered to have nodal disease if they had positive SLNB or developed nodal disease after negative or failed SLNB. Patients were in the no nodal disease group if they had negative or experienced SLNB failure with no regional nodal disease during the follow-up period. Patients with failed or negative SLNB who received adjuvant radiation therapy to the nodal basin were excluded from the no nodal disease group.

The SLNB failure rate in this study cohort (29.0%) exceeds that reported for HN melanoma and cutaneous squamous cell carcinoma (approximately 2%).^{29,30} Prior HN MCC studies have rarely reported SLNB failure rates, although a small series found a rate of 25%, consistent with our findings.¹³ Failures were most often due to radiotracer nonvisualization, a known challenge in the HN region.¹⁵⁻¹⁷ MCC-specific tumor biology may also contribute: we observed higher failure rates in patients with high-risk features (eg, LVI, deep invasion, size >2 cm, thickness >10 mm) but may not have been adequately powered to observe a moderate or large effect size. LVI and tumor emboli could disrupt lymphatic flow, complicating mapping.^{31,32} Additionally, surgeon experience and a learning curve with performing SLNB, as demonstrated in the MSLT-1 melanoma trial,³³ could influence the high failure rates observed in our study.

Tumor location had a large effect on SLNB failure, while female sex and older age had moderate effect. Tumor site differences may relate to what is termed *shine-through*, an overlap of radioactivity from the primary site and the SLN,³⁴ particularly in the cheek and neck. Older age has been associated with higher nonvisualization rates in melanoma and breast cancer, likely due to age-related anatomic changes.³⁵⁻³⁸ Techniques, such as SPECT/CT or use of radiotracers such as 99mTc-tilmanocept or indocyanine green-99mTc-nanocolloid, may help mitigate these challenges,³⁹⁻⁴³ and could be explored in future work. In our study, we observed a similar rate of SLNB failure with SPECT/CT, but this conclusion is limited by the frequency of SPECT/CT use (12 of 69 participants [17.4]).

Overall, patients with aggressive tumor characteristics and/or tumors in the cheek and neck may warrant counseling regarding the possibility of SLNB failure and consideration of preoperative and postoperative imaging. Our findings add to growing evidence of the challenges associated with SLNB in the HN, which may be secondary to anatomic variability, inconsistent lymphatic drainage, and both technical and biologic factors.^{7,8,44-46} Better preoperative predictors of nodal disease are needed.

In our study, LVI, infiltrative growth pattern, and tumor size had the largest effect sizes on nodal disease, while tumor thickness and age had smaller effect sizes on nodal disease,

supporting their use in risk stratification.^{4,47-51} Patients of older age or with tumors greater than 1 cm, thickness greater than 4 mm, LVI, or infiltrative growth pattern may benefit from intensified follow-up, early or more frequent imaging, and/or strong consideration of nodal basin radiation therapy regardless of SLNB outcome. Notably, only 1 of 9 patients treated with aRT to the nodal basin after failed or negative SLNB experienced recurrence, and it was distant recurrence.

Additionally, unlike past literature reporting HN SLNB outcomes for predominantly patients treated with WLE,⁸ our study cohort uniquely consisted of patients mostly treated with MMS (55 patients [79.7%]), with high rates of staging with SLNB (62 of 73 patients [84.9%]) and/or pre-MMS staging imaging (36 of 73 patients [49.3%]) in the full HN cohort. Our work highlights the potential of MMS as a tissue-sparing treatment modality for HN MCC tumors that does not compromise performance of SLNB or presurgery imaging.

Limitations

Limitations include retrospective study design, potential underreporting of histologic features, and incomplete follow-up for a small number of patients. Although our single-institution SLNB cohort is among the larger series in HN MCC, sample size remains limited for detecting some associations. Fewer T2 and T3 tumors were included (tumor size > 2 cm, 10%), possibly due to careful exclusion of patients with clinically evident nodal disease. Merkel cell polyomavirus status was unavailable for most cases; incorporating this variable in future work may help identify patients at greater risk for SLNB positivity or failure. Surgeon experience was not measured and may have influenced SLNB failure, although we did not observe differences in SLNB failure rates between cases done before 2020 and after 2020. Our high rates of failed SLNB (20 of 69 patients [29.0%]) may also skew our characterization of SLNB accuracy.

An additional limitation relates to our definition of FN SLNB. We defined FN as regional nodal recurrence as the first site of recurrence without concurrent local or in-transit recurrence.^{7,17,19,20} By this definition, a nodal recurrence could occur in a basin not sampled by the sentinel lymph node biopsy, rather than truly representing a missed sentinel node

metastasis. As a result, the FN rate we found may differ from studies that define FN more narrowly (eg, counting only same-basin recurrences), and direct comparison across studies should be interpreted with caution.

Conclusions

This cohort study found that SLNB remains a valuable staging tool in HN MCC, identifying more than half of patients with

occult nodal disease and guiding treatment decisions. Its performance, however, is reduced in this anatomically complex region, with notable rates of failure and FN results. Patients with certain tumor sites and/or with high-risk pathologic features (eg, LVI, infiltrative growth pattern, and tumor size) may especially benefit from tailored management—ie, early surveillance imaging and/or consideration of nodal basin radiation—even when SLNB is negative. These findings support both the continued use of SLNB and the need for adjunctive strategies to optimize nodal staging in HN MCC.

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